BHAWUK PAHUJA

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Summary

Player Engineer / Frontend Engineer with 3+ years building high-performance HTML5 streaming players across Smart TV and browser platforms. Deep hands-on expertise with **Shaka Player**, DASH & HLS, DRM, and adaptive bitrate (ABR) streaming, backed by strong React engineering. Proven at debugging complex live/DVR playback issues, driving root-cause analysis on production incidents, and shipping resilient playback UX on webOS (LG), Tizen (Samsung), and desktop browsers.

Core Skills

Player & Streaming:	Shaka Player, HTML5 Video, DASH, HLS, Adaptive Bitrate (ABR), Live & DVR, Manifest/Segment handling, Trickplay/Thumbnails, Subtitles
DRM & Media:	Widevine, PlayReady, Encrypted Media (EME), Picture-in-Picture (Document PiP API)
Frontend:	React.js, JavaScript (ES6+), Redux, HTML5, CSS3, SCSS, Component Libraries, Responsive UI
Platforms & Tools:	Smart TV (webOS / Tizen), Browser, REST APIs, Git, Chrome DevTools, Performance Profiling

Work Experience

Engineer – Software Development **Oct 2024 – Present**

Xperi (TiVo) — Pune, India *Key Achievements*

- **Owned Shaka Player modules** across TiVo Plus web and Smart TV apps, tuning ABR, buffering, and manifest-refresh behavior for smooth live, DVR, and VOD playback.
- **Resolved critical playback failures** (e.g., Shaka error 7000 / segment-fetch and manifest edge cases), cutting user-facing interruptions and stalls on live/DVR streams.
- **Shipped Picture-in-Picture** using browser-native and Document PiP APIs, consolidating scattered PiP logic into a single robust, production-grade module.
- **Drove root-cause analysis** on high-priority production playback incidents, coordinating with NOC and product teams to reproduce, triage, and resolve device-specific failures.
- **Built trickplay thumbnail previews** for the seek bar, handling image sprites and aspect-ratio rendering across web and Smart TV players.
- **Added multi-audio and language-track switching** in the player, wiring track selection UI to Shaka's audio/text track APIs for a seamless viewer experience.
- **Improved player-UI performance by ~30%** by optimizing DOM manipulation and React rendering across Smart TV and browser platforms.
- **Boosted dev velocity by ~25%** by building reusable player-facing React component libraries and reducing UI code duplication.

Associate Software Developer **Jul 2023 – Oct 2024**

Xperi (TiVo) — Pune, India *Key Achievements*

- **Enhanced core video playback flows**, improving stream stability and reducing interruptions across webOS (LG) and Tizen (Samsung) devices.
- **Fixed complex buffering, rendering, and UI-crash issues** unique to constrained Smart TV runtimes.
- **Delivered performance-critical player-UI features** focused on render efficiency, memory footprint, and smooth D-pad navigation.
- **Hardened playback error handling** with graceful recovery and retry logic to reduce fatal playback exits on unstable networks.
- **Instrumented playback analytics** to surface buffering, startup-time, and error metrics that guided player reliability improvements.
- **Resolved cross-platform quirks** such as localStorage persistence on LG webOS hosted apps to keep player state and preferences consistent across devices.

SDE Intern **Feb 2022 – Jul 2023**

Xperi (TiVo) — Pune, India *Key Achievements*

- **Delivered multiple front-end features** for core Smart TV app workflows; built reusable React components and contributed to code reviews and clean-architecture practices.
- **Improved navigation responsiveness** on low-powered TV hardware through targeted rendering optimizations.

Frontend Engineering Highlights

- **Architected reusable React components** and design-system primitives that standardized UI across multiple streaming apps.
- **Applied advanced React performance techniques** — memoization, virtualized lists, and render-cycle optimization — to sustain 60fps navigation on low-powered Smart TV hardware.
- **Built responsive, accessible player UI** with focus management and keyboard/D-pad navigation for seamless browser and Smart TV experiences.

Education

B.E., Computer Science Engineering — Chitkara University **2019 – 2023**

GPA: 9.93 / 10